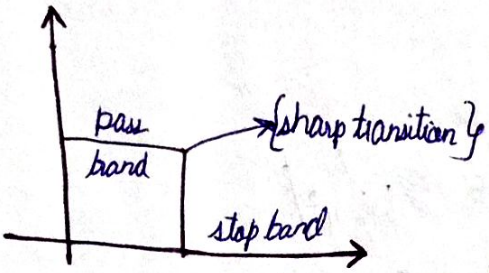
Microwave filter



$\rightarrow$ LPF, HPF, BPF, BSF.

In practical filter;

Gradual transition from :-
Passband to stopband or vice versa.

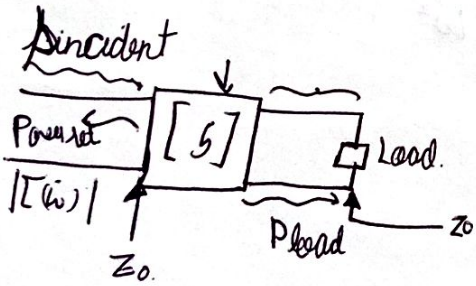
Approximated
- Butterworth / Binomial / Equiripple / flat response.
- Chebyshev (1'st kind)
- Chebyshev (2nd kind)
- elliptical filter
- Bessel filter / Linear phase filter.

} ripple & roll off is good

Insertion loss method :-

(to design Microwave filter)

Power loss ratio (PLR) = $\dfrac{\text{Power available from the source}}{\text{Power delivered to the load}}$



$$PLR = \frac{P_{inc}}{P_{inc} - P_{ret}} = \frac{1}{1 - \frac{P_{ret}}{P_{inc}}} = \frac{1}{1 - |\Gamma(\omega)|^2}$$

$$|\Gamma(\omega)|^2 = \left| \frac{v_{ret}}{v_{inc}} \right|^2$$

$$|S_{11}|^2 + |S_{21}|^2 = 1 \rightarrow \text{lossless.}$$

$$|S_{21}|^2 = 1 - |S_{11}|^2$$

For matched cond^n at both ports, $|\Gamma(\omega)|$ can be written as $|S_{11}|$

$$P_{LR} = \frac{1}{1-|S_{11}|^2} = \frac{1}{|S_{21}|^2}.$$

$$\text{Insertion loss } (IL) = 10 \log P_{LR} = 10 \log \left\{\frac{1}{|S_{21}|^2}\right\}.$$

$$= \boxed{-20 \log |S_{21}|}$$

It can be shown that $|\Gamma(\omega)|^2$ is an even f^n of 'ω'. So, $|\Gamma(\omega)|^2$ is an even f^n of ω $\therefore$ a polynomial in ω^2

$$|\Gamma(\omega)|^2 = \frac{M(\omega^2)}{M(\omega^2)+N(\omega^2)}$$

$$P_{LR} = \frac{1}{1-|\Gamma(\omega)|^2} = \frac{1}{1-\frac{M(\omega^2)}{M(\omega^2)+N(\omega^2)}}$$

$$P_{LR} = 1 + \frac{M(\omega^2)}{N(\omega^2)}$$

where M & N are real polynomial in ω^2.

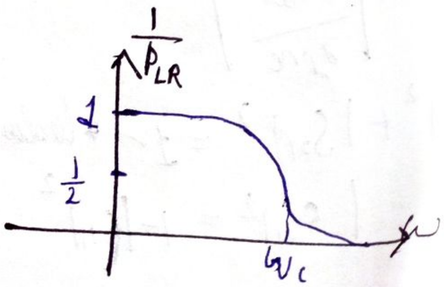
Maximally flat / Butterworth filter

$$P_{LR} = 1 + k^2 \left(\frac{\omega}{\omega_c}\right)^{2N}$$

$k \to$ ripple factor
$\omega \to$ any angular freq.
$\omega_c \to$ angular cutoff freq.
$N \to$ Order of filter.

if $k = 1$ & $\omega = \omega_c$

$$P_{LR} = 1 + (1)^{2N} = 2$$

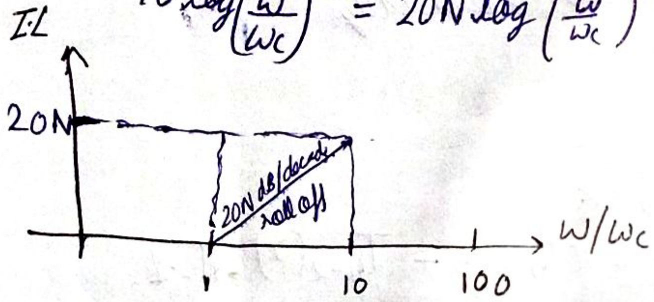


For larger ω & $K=1$,

$$P_{LR} = 1+\left(\frac{\omega}{\omega_c}\right)^{2N} = \left(\frac{\omega}{\omega_c}\right)^{2N}$$

$$IL = -20\log|S_{21}| = -20\log\frac{1}{\sqrt{P_{LR}}} = 10\log P_{LR}$$

$$= 10\log\left(\frac{\omega}{\omega_c}\right)^{2N} = 20N\log\left(\frac{\omega}{\omega_c}\right)$$



__Microwave filter:-__

__Insertion loss__: $IL(dB) = 10\log P_{LR}$

@ Butterworth filter, $P_{LR} = 1+K^2\left(\frac{\omega}{\omega_c}\right)^{2N}$

$\omega > \omega_c$; $P_{LR} \approx K^2\left(\frac{\omega}{\omega_c}\right)^{2N}$

for $K^2 = 1$, {Ripple factor}

$P_{LR} = \left(\frac{\omega}{\omega_c}\right)^{2N}$

$$\Rightarrow IL = 10\log\left(\frac{\omega}{\omega_c}\right)^{2N}$$

$$\frac{IL}{10} = 2N\log\left(\frac{\omega}{\omega_c}\right)$$

now, given $f_c = 400MHz$

@ $f = 1GHz$

$IL = 20dB$

$$\boxed{N = \frac{\log\left(10^{IL/10}\right)}{2\log\left(\frac{\omega}{\omega_c}\right)}}$$

Chebyshev filter

$P_{LR} = 1 + k^2 T_N^2\left(\frac{\omega}{\omega_c}\right)$

$T_N(x) \rightarrow$ Poly factor : $\underbrace{\text{is a a chebyshev poly}}$ ⟨ 1st Kind / 2nd Kind

oscillatory f^n for $x \in [-1, 1]$ beyond which, it increases monotonically.

1st Kind Chebysev Polynomial:

$$T_0(x) = 1$$
$$T_1(x) = x$$
$$T_N(x) = 2x T_{N-1}(x) - T_{N-2}(x), N > 1$$

@ $N = 2$,

$$T_2(x) = 2x^2 - 1$$
$$T_3(x) = 2x(2x^2 - 1) - x = 4x^3 - 3x$$
$$T_4(x) = 2x(4x^3 - 3x) - (2x^2 - 1) = 8x^4 - 8x^2 + 1$$

and so on.

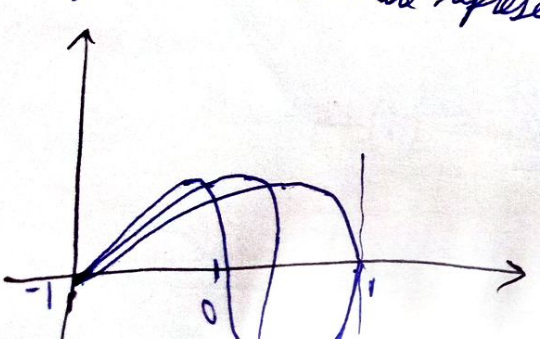
diff. between the first & second kind only exists for $T_1(x)$ that is, $T_1(x) = 2x$ for second kind.

@ $-1 < x < 1$

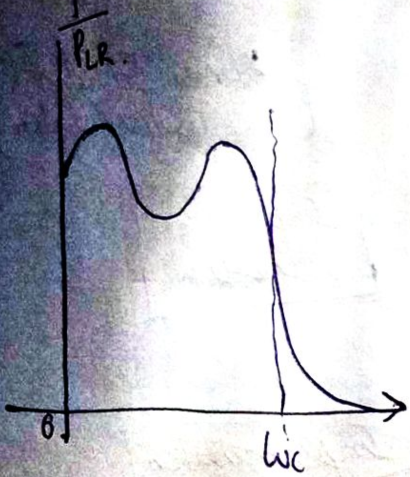
$$T_0(x) = 1$$
$$T_1(x) = [-1, 1]$$
$$T_2(x) = [-1, 1]$$
$$T_3(x) = [-1, 1]$$

We will always get $[-1, 1] \rightarrow$ oscillating values.

Chebysev $\rightarrow$ can also be represented trignometrically.

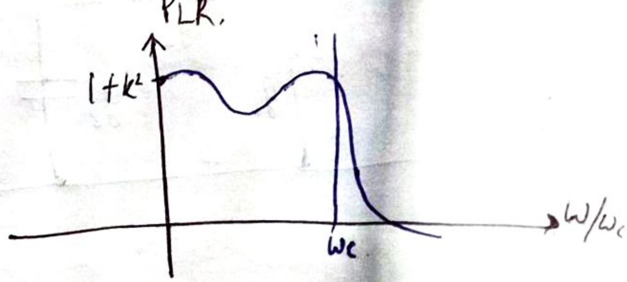


Can be anything like th...



← we did not consider the $-$ve frequency. we are concerned only about the +ve one.

order	P_{LR}
2nd / $N=2$	$1+k^2T_2^2(\omega/\omega_c)$ $=1+k^2\left(2\left(\frac{\omega}{\omega_c}\right)^2-1\right)^2$ @ $\omega=0$, $P_{LR}=1+k^2$ @ $\omega=\omega_c$, $P_{LR}=1+k^2$
$N=3$	$1+k^2\left(4\left(\frac{\omega}{\omega_c}\right)^3-3\left(\frac{\omega}{\omega_c}\right)\right)^2$



→ @ even order filter, $\omega=0$ & $\omega=\omega_c$
→ $P_{LR}=1+k^2$ & @ odd filter
$\omega=0 \Rightarrow P_{LR}=1$

for ω is large

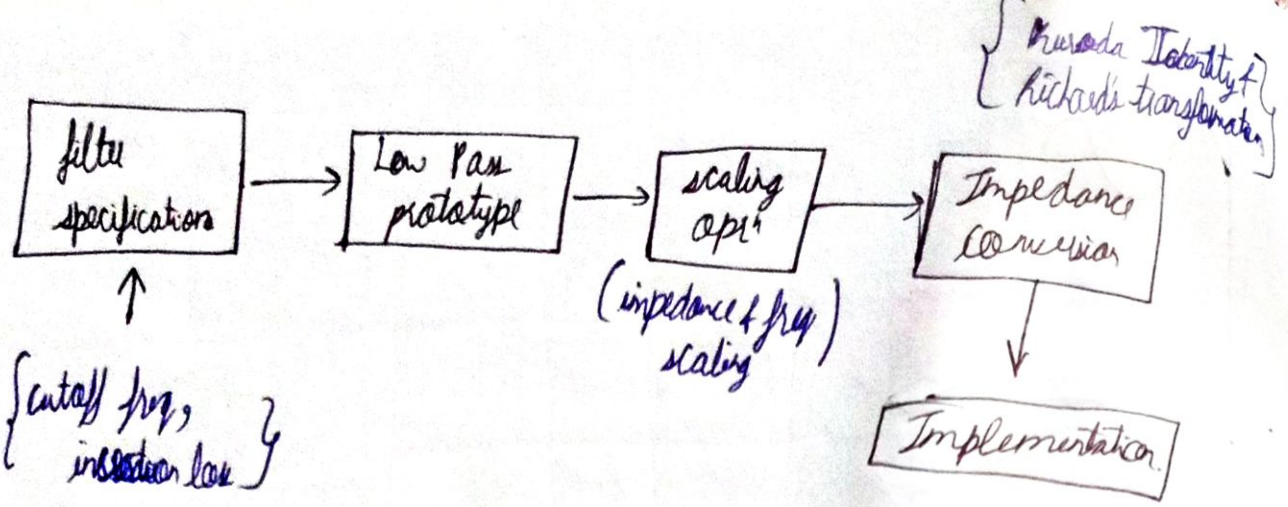
$$P_{LR}\approx\frac{k^2}{4}\left(\frac{2\omega}{\omega_c}\right)^{2N} \rightarrow \text{chebyshev}.$$

@ Butterworth, $P_{LR}\approx k^2\left(\frac{\omega}{\omega_c}\right)^{2N}$ @ ω is large.

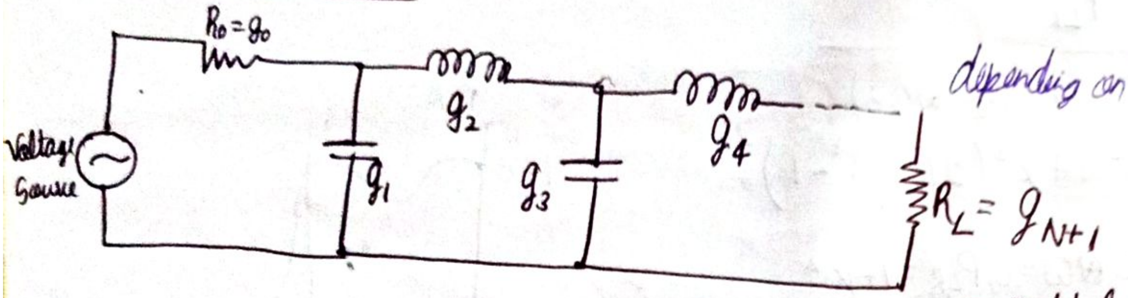
P_{LR} is more in Chebyshev by a factor of $2^{2N}/4$ → IL in chebyshev = $10\log\frac{2^{2N}}{4}$ dB more than BW.

$$\boxed{N=\frac{\log 10^{IL/10}\times 4/k^2}{2\log(\omega/\omega_c)}}$$ ★★

General procedure to design a filter:

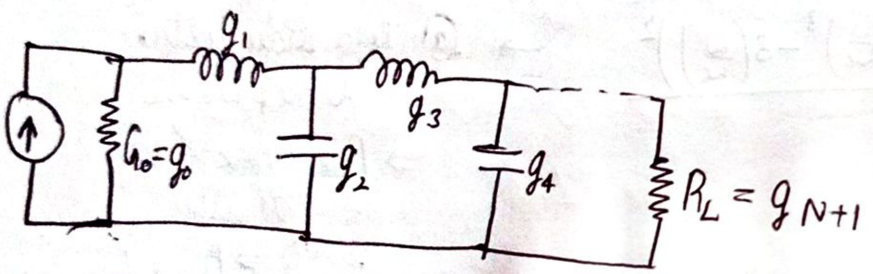


Realizing LPF prototype



depending on order of filter

→ 1st element = shunt capacitor



→ 1st element = series inductor

g → generator, g_0 → source impedance

N → order of the filter g_{N+1} → load imp

we get g_1, g_2, g_3 and so on ......

$R_s' = 1 \cdot R_0 = R_0$

$R_L' = 1 \cdot R_0 = R_0$

Or sometimes (when $R_L \neq 1$)

$R_L \cdot R_0 = R_L \cdot R_0$

② Frequency scaling:-

Here we replace ω by $\frac{\omega}{\omega_c}$

$$X_L = \omega L$$

$$X_L' = \frac{\omega L}{\omega_c} = \omega L'$$

$$\therefore L' = \frac{L}{\omega_c}$$

$$L' = \frac{g_k}{\omega_c}$$

$$X_C = \frac{1}{\omega C}$$

$$X_C' = \frac{\omega_c}{\omega \cdot C} = \frac{1}{\omega C'}$$

same

$$\therefore C' = \frac{C}{\omega_c}$$

$$C' = \frac{g_k}{\omega_c}$$

Applying Impedance + freq. scaling both:

$$\boxed{L_k' = \frac{L_k R_o}{\omega_c} \qquad C_k' = \frac{C_k}{R_o \omega_c}}$$

Finding g-parameter for butterworth LPF prototype:-

$$g_0 = g_{N+1} = 1 \text{ (normalized value)}$$

$$g_k = 2 \sin\left[\frac{(2k-1)\pi}{2N}\right], k = 1, 2, 3, \ldots N)$$

	g_0	g_1	g_2	g_3	g_4	g_5
N = 1	1	2	1			
N = 2	1	1.414	1.414	1		
N = 3	1	1	2	1	1	
N = 4	1	0.7658	1.847	1.847	0.7654	1

Impedance scaling

→ multiply impedance by R_0 (characteristic resistance)

$$X_L = \omega L$$

Applying impedance scaling

$$X_L' = \omega L R_0 = \omega L'$$

$\therefore L' = L R_0$ (new inductance after impedance scaling)

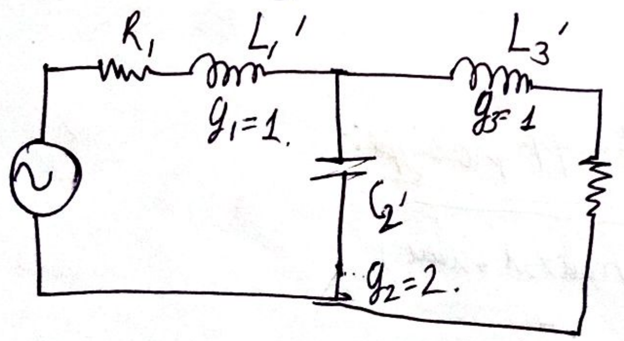
$$= g_k R_0$$

$$X_c = \frac{1}{\omega C}$$

$$X_c' = \frac{1}{\omega C} \times R_0$$

$$\Downarrow$$

$$\frac{1}{\omega C'} = \frac{1}{\omega\left(\frac{C}{R_0}\right)} \quad \therefore \boxed{C' = \frac{C}{R_0}} \quad C' = \frac{g_k}{R_0}$$



given: $R_0 = 50\,\Omega$

$f_c = 3\,GHz$

g parameters

(N=3)

$$L_k' = \frac{L_k R_0}{\omega_c} \qquad C_k' = \frac{C_k}{R_0 \omega_c} \Rightarrow$$

$L_1' = (g_1 \times 50)/2\pi \times 3\times10^9$
$= 2.653\times10^{-9}\ H$

$C_2' = (g_2 \times 1)/50\times2\pi\times3\times10^9$
$= 2.123\times10^{-12}\ F$

$L_3' = (g_3 \times 50)/2\pi\times3\times10^9$
$= 2.653\times10^{-9}\ H.$

Filter Implementation [Richard's transformation]

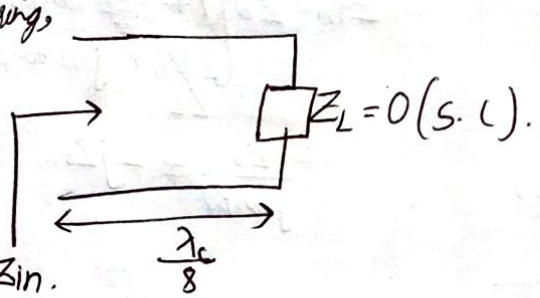
$$Z_{in} = Z_0\left[\frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan\beta l}\right]$$

L

impedance, $Z = j\omega L$

If $l = \frac{\lambda_c}{8}$

Considering,



equating the impedances,

$$jZ_0 = j\omega_c l$$

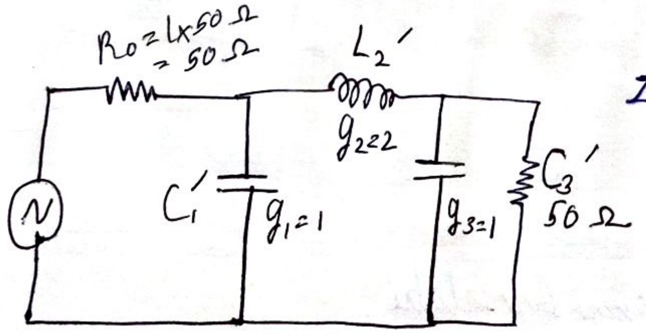
$\boxed{Z_0 = \omega_c L}$ (to get the characteristic impedance)

$Z_L = 0,\ Z_{in} = jZ_0 \cancel{\tan\beta l}$

$Z_{in} = jZ_0$ (purely inductive)

Design a maximally flat filter → another name of Butterworth.

$f_c = 400\ MHz$

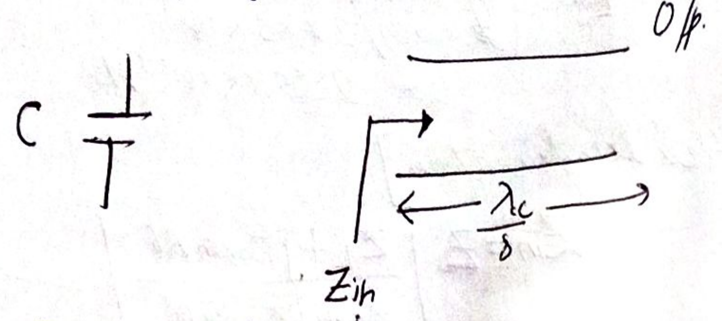


IL @ 1 GHz = 20 dB, $R_0 = R_L = 50\ \Omega$
(we found in last class order of filter = 3)

$$C_1' = \frac{g_1}{R_0 \omega_c} = \frac{1}{50\times2\pi\times400\times10^6} = 7.95\ pF$$

$$L_2' = \frac{g_0 R_0}{\omega_c} = \frac{1}{8\pi \times 10^6} = 39.8\,\mu H$$

$$C_3 = \frac{g_3}{R_0 \omega_c} = 7.95\,pF$$



$$Z = \frac{1}{j\omega C} = \frac{-j}{\omega C}$$

$$Z_L = \infty$$

$$Z_{in} = Z_0 \left[\frac{1 + j\frac{Z_0}{Z_L}\tan\beta l}{\frac{Z_0}{Z_L} + j(1)\tan\beta l}\right]$$

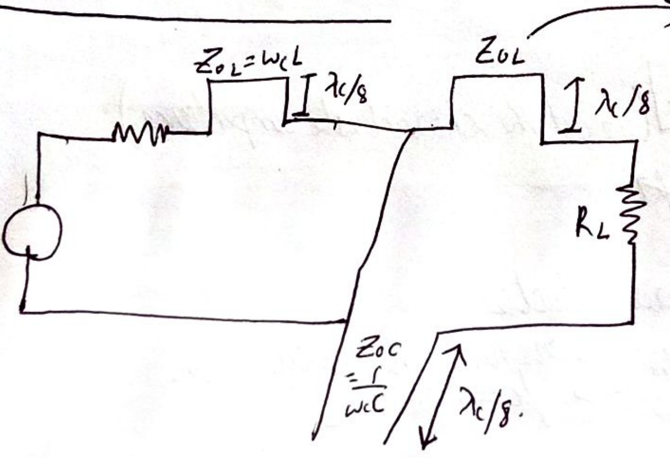
equating,

$$= \frac{Z_0}{j\tan\beta l} = -jZ_0.$$

$$\frac{-j}{\omega_c C} = -jZ_0$$

$$\therefore \boxed{Z_0 = \frac{1}{\omega_c C}}$$

Realizing the prev. network



→ for chebyshev we would have had diff. g parameter and hence diff. values of Z_{0L} for diff. inductors

Kuroda Identity

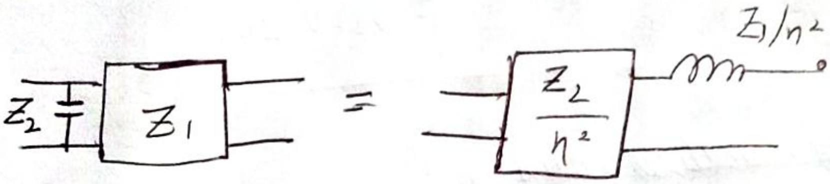
Advantages:

→ It physically separates transmission line stubs.

→ Transform series stub to shunt stub & vice versa.

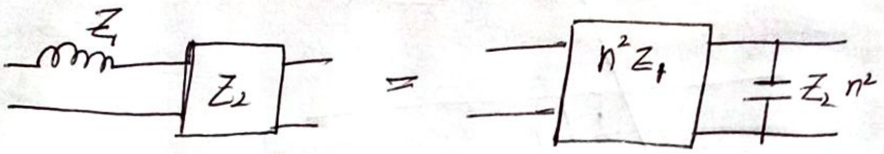
→ Change impractical characteristic impedance into more realizable impedance

1st Kuroda Identity:



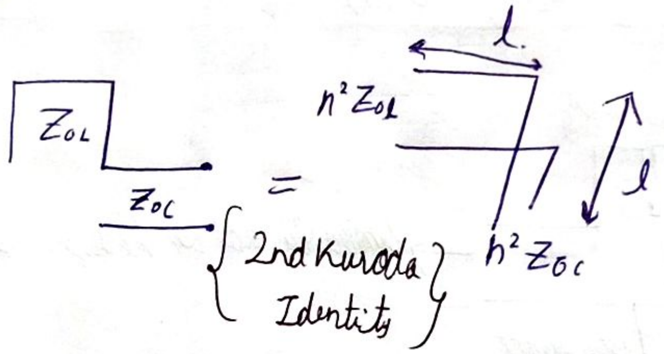
where $n^2 = 1 + \frac{Z_2}{Z_1}$

2nd Kuroda Identity:



$n^2 = 1 + Z_2/Z_1$

in the prev. n/w, we have

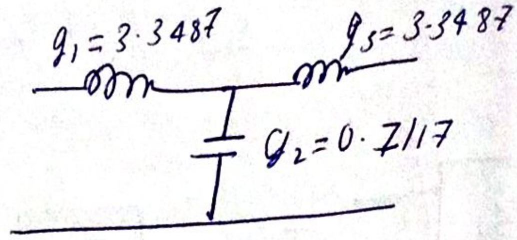


Chebyshev / Equal ripple filter

N	g_1	g_2	g_3
1	-	-	-
2	-	-	-
3	3.3487	0.7117	3.3487

Problem 1- Design a 3dB ripple low pass filter using microstrip line...
Specifications are :
cutoff freq. = 4 GHz
Order of the filter = 3
Characteristic impedance
@ source & load = 50 Ω
(R_s & R_L)

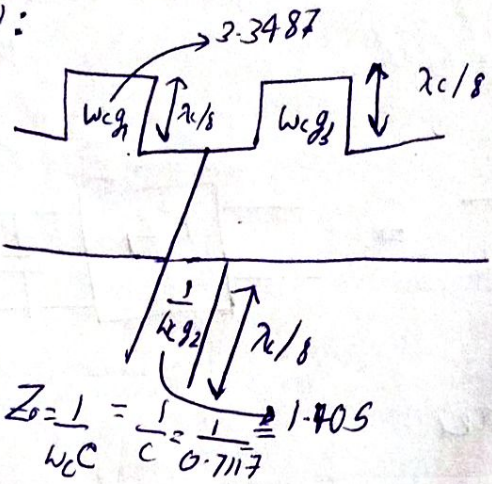
step 1:- Find g parameter for



(we are using normalized value in)

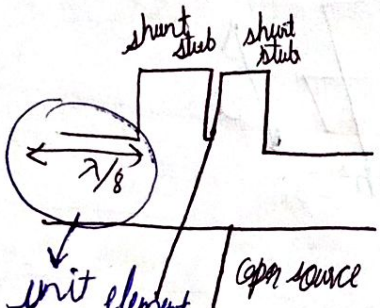
step 2:- Realize the n/w:

$W_c = 1$ (normalized)



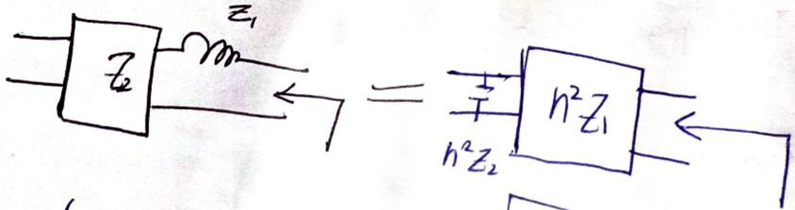
$Z_0 = \frac{1}{W_c C} = \frac{1}{C} = \frac{1}{0.7117} = 1.405$

step-3:



→ similarly add on the right side

Step-4: Apply 2nd kuroda identity



$$\boxed{n^2 = 1 + \frac{Z_2}{Z_1}}$$

(trans) $n^2 Z_1 = (1.298)(3.3487) \cdot 4.35$

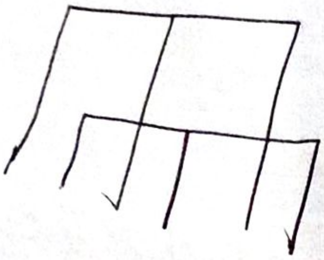
$(Cap)_1$ $n^2 Z_2 = (1.298)(1) = 1.298$

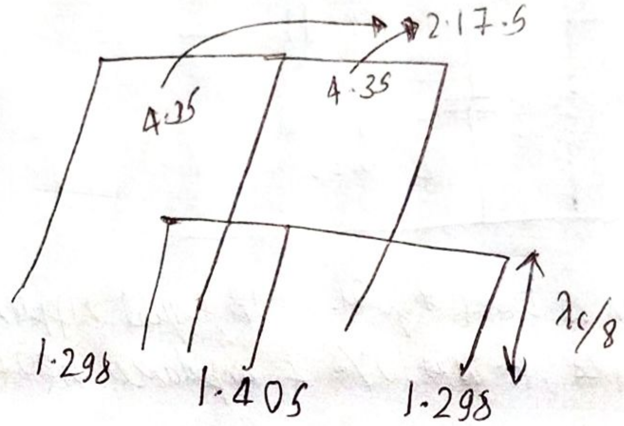
$n^2 = 1 + \frac{1}{3.3487}$
$= 1.298$.

similarly for second stub.

trans → 4.35 and

Cap_2 → 1.298





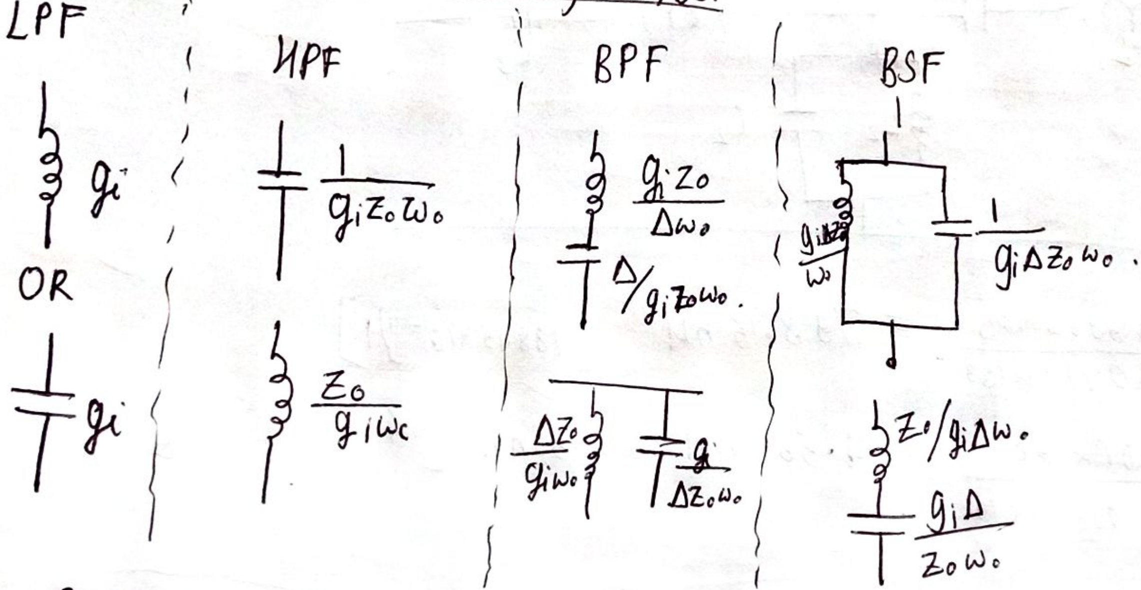
$$\lambda_c = \frac{c}{f} = \frac{3\times10^8}{4\times10^9}$$

$$= 7.5\,\text{cm}$$

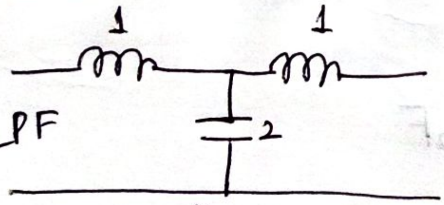
⇓
multiplying the imp. to find Correct value, $L_1 = 1.298\times50$
$= L_3 = 64.9\,\Omega$
$C_2 = 1.405\times50$
$= 70.3\,\Omega$.

$\boxed{\text{Inductal} \rightarrow \text{S.T.L}}$

Transformation from LPF to HPF, BPF, BSF



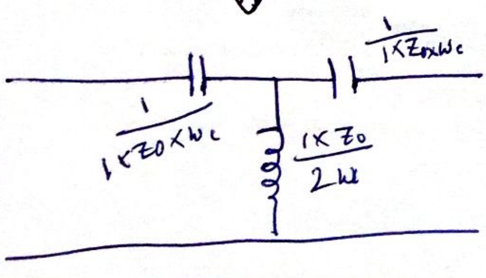
eg - 3rd order:

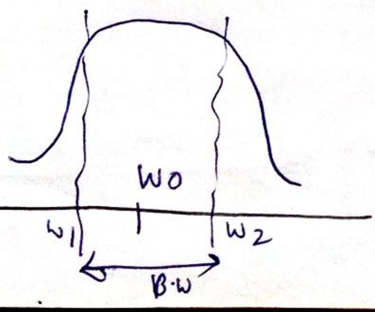


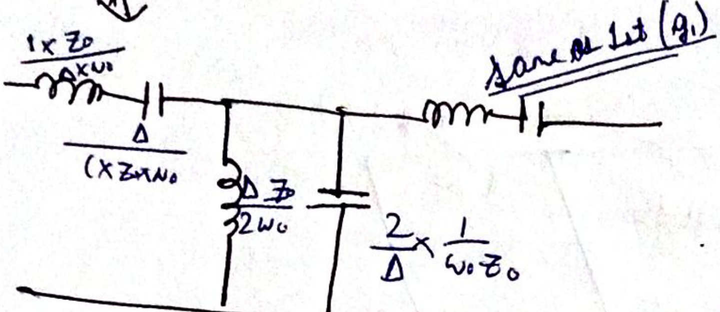
$$\Delta = \frac{\omega_2-\omega_1}{\omega_0}$$

$$\omega_0 = \sqrt{\omega_1\omega_2}$$

⇓

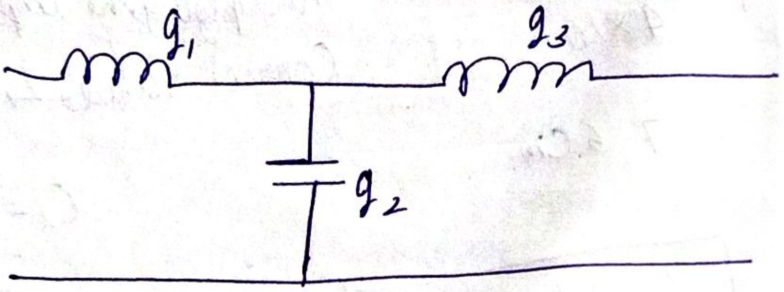




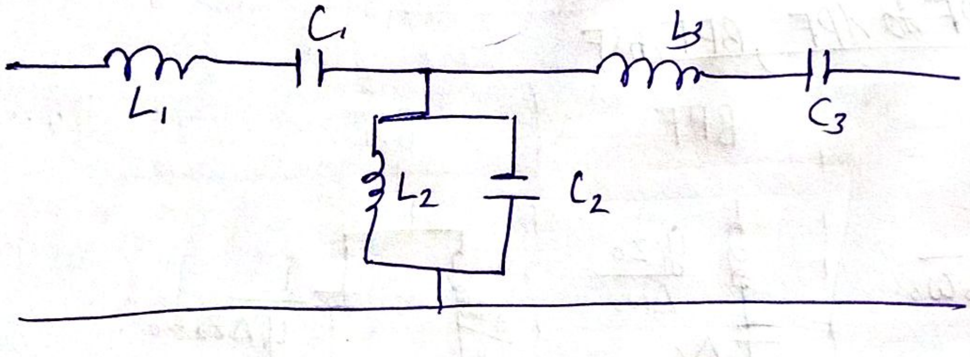


Q - Design a bandpass filter. 0.5 dB equal ripple response with order of 3. The c... is 1 GHz, B.W is 10% & impedance is 50 Ω.

Given, $g_1 = 1.5963$, $g_2 = 1.0967$, $g_3 = 1.5963$



⇓



$$L_1 = \frac{1.5963 \times 50}{(0.1) \times 1 \times 10^9} = 798.15 \text{ nH} \quad [798.15 \times 10^{-9} H]$$

$$L_2 = \frac{(0.1) \times 50}{1.0967 \times 1 \times 10^9} = 4.559 \text{ nH} \approx 4.56 \text{ nH} \quad [4.56 \times 10^{-9} H]$$

$$C_1 = \frac{(0.1)}{1.5963 \times 50 \times 1 \times 10^9} = 1.252 \text{ pF} \quad [0.00125 \times 10^{-9} F]$$

$$C_2 = \frac{1.0967}{(0.1)(50)(10^9)} = 0.2193 \text{ nF} = 219.3 \text{ pF}$$